

# Nick Kunz

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## Research Interests

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My research interests are at the intersection of data science, machine learning, and complex systems. Many of the question I ask are motivated by understanding the emergence of system dynamics within the context of evolving real-world constraints. My work has direct applications in urban planning, economics, defense, and other areas of societal impact.

## Education

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| <b>Cornell University</b> — Ithaca, NY<br>Ph.D., Systems Engineering, Artificial Intelligence<br><i>Advisors: H. Oliver Gao, Francesca Parise, Wenfei Xu</i>                            | 2027 |
| <b>Cornell University</b> — Ithaca, NY<br>M.S., Systems Engineering<br><i>Advisors: H. Oliver Gao</i>                                                                                   | 2025 |
| <b>Columbia University</b> — New York, NY<br>M.S., Urban Planning, Urban Analytics<br><i>Advisors: Lance Freeman, Kazuki Sakamoto</i>                                                   | 2019 |
| <b>Harvard University</b> — Cambridge, MA<br>Urban Planning<br><i>Advisor: Nick Hornig</i>                                                                                              | 2012 |
| <b>Arizona State University</b> — Tempe, AZ<br>B.S. (Hons.), <i>summa cum laude</i> , Housing and Urban Development<br><i>Advisors: Kevin Kellogg, Michael Underhill, Trevor Barger</i> | 2012 |

## Publications

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### Journal Articles

- N. Kunz, H. O. Gao, “A General Capacity Frontier of Complex Networks”. *Preprint arXiv:2610.09574*. 2026. [\[Link\]](#)
- N. Kunz, H. O. Gao, “Global Geolocated Realtime Data of Interfleet Urban Transit Bus Idling”. *Preprint arXiv:2403.03489*. 2026. [\[Link\]](#)
- D. Lu, L. Cui, N. Kunz, et. al., “Urban Infrastructure and Spatiotemporal Environmental Features for EGFR-Mutant Lung Cancer”. *Preprint medRxiv:18.26353481v1*. 2026. [\[Link\]](#)
- C. Avci, C. Huang, N. Kunz, H. O. Gao, “uTECH-Population: Synthetic Population Generation with Housing Assignment”. *Under Review*. 2026.

### Conference Proceedings

- W. Wu, N. Kunz, P. Branco, “ImbalancedLearningRegression” in *Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD, September 19–23, 2022, Grenoble, France*, Proceedings, Part VI, Lecture Notes in Computer Science, vol 13718. Springer, pp. 645–648, 2023. [\[Link\]](#)

### Working Papers

- N. Kunz, H. O. Gao, “Global Behavioral Patterns of Interfleet Urban Transit Bus Idling”. *In Preparation*. 2026.
- N. Kunz, H. O. Gao, “Network Constrained Spatial Prediction: Transfer Learning Geolocations”. *In Preparation*. 2026.

## Theses

N. Kunz, “Structural Systems Intelligence: Network-Constrained Estimation of Global Transit Dynamics,” Ph.D. Dissertation, Duffield College of Engineering, Cornell Univ., Ithaca, NY, 2027.

N. Kunz, “Unsupervised Learning for Submarket Modeling: A Proxy for Neighborhood Change,” M.S. Thesis, Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available: Columbia Academic Commons. [\[Link\]](#)

N. Kunz, “Realizing Interactive Architecture: A Driver of the Knowledge Economy,” B.S. (Hons.) Thesis, Barrett, The Hons. Col., Arizona State Univ., Tempe, AZ, 2012. Available: Barrett, The Honors College Thesis/Creative Project Collection. [\[Link\]](#)

## Industry Reports

N. Kunz, “ $n$ -Dimensional Mapping by Normalized Value Approximation: Squad Immersive Virtual Trainer (SiVT) Safety Performance Metrics,” Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2021. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

N. Kunz, “Normalized  $n$ -Dimensional Inverse Euclidean Distance: Combining Independent Performance Metrics,” Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2021. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

N. Kunz, “Minimum-Weighted Mean Variability: Within Group Performance Metrics,” Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2020. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

N. Kunz, “An Exploratory Data Analysis of the Gavin Cup: Bringing Readiness & Performance Metrics Into Focus,” Integrated Visual Augmentation System (IVAS), Microsoft, Redmond, WA, 2020. Protected from public disclosure pursuant to: Trade Secrets Act 18 USC § 1905, 5 USC § 552(b)(4), and C3.2.1.4. of US Dept. of Defense Regulation 5400.7-R. DISTRIBUTION F.

## Software Packages

### GRD-TRT-BUF-4I: Ground Truth Buffer for Idling

2023 - 2027

 [Github](#) [ [Link](#) ]  [arXiv](#) [ [Link](#) ]  [Website](#) [ [Link](#) ]

A novel algorithm designed as a series of scalable microservices that detects urban transit bus idling in realtime on global scale. Utilizes a combination of templated ETL pipelines to provide a comprehensive view of the emergent behavior of interfleet idling patterns across multiple transit networks around the world.

### SMOBN: Synthetic Minority Over-Sampling Technique for Regression with Gaussian Noise

2021

 [Github](#) [ [Link](#) ]  [PyPI](#) [ [Link](#) ]  [Kaggle](#) [ [Link](#) ]  [Stack Exchange](#) [ [Link](#) ]

A pre-processing algorithm designed to address imbalanced data for regression problems. Conducts over-sampling with traditional interpolation, as well as with the introduction of Gaussian noise. Selects between the two over-sampling techniques by the KNN distances underlying a given observation.

### NestedHyperBoost: Nested Cross-Validation for Bayesian Optimized Gradient Boosting

2020

 [Github](#) [ [Link](#) ]  [PyPI](#) [ [Link](#) ]

Unifies Nested K-Fold Cross-Validation, Bayesian Hyperparameter Optimization, and Gradient Boosting. Designed for rapid prototyping on small to mid-sized data sets. Quickly obtains high quality prediction results by abstracting away tedious hyperparameter tuning and implementation details in favor of usability and implementation speed.

### NestedHyperLine: Nested Cross-Validation for Bayesian Optimized Linear Regularization

2020

 [Github](#) [ [Link](#) ]  [PyPI](#) [ [Link](#) ]

Unifies Nested K-Fold Cross-Validation, Bayesian Hyperparameter Optimization, and Linear Regularization. Designed for rapid prototyping on small to mid-sized data sets. Quickly obtains high quality prediction results by abstracting away tedious hyperparameter tuning and implementation details in favor of usability and implementation speed.

 Github [ [Link](#) ]  CRAN [ [Link](#) ]

A novel matrix sampling algorithm designed to conditionally random sample sparse matrices. Useful for training and test set splitting prior to model fitting in cross-validation procedures and for estimating the predictive accuracy of imputation methods, such as NNMF and SVD.

## Selected Projects

N. Kunz, S. Huang, S. Osakata, “Cloudy Night: Activating Park Space by Illuminated Interactivity,” Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available:

 GSAPP [ [Link](#) ]  Github [ [Link](#) ]  YouTube [ [Link](#) ]  Faculty [ [Link](#) ]

N. Kunz and S. Bhele, “Signal or Noise? Qualified Opportunity Zone Affect on Development Activity in Queens,” Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2019. [Online]. Available:

 GSAPP [ [Link](#) ]  Github [ [Link](#) ]  Issuu [ [Link](#) ]

E. Hudgins and N. Kunz, “Planning for Extreme Settlements, McMurdo Station, Antarctica,” Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2018. [Online]. Available:

 GSAPP [ [Link](#) ]  Issuu [ [Link](#) ]

J. Karasik, S. Lee, N. Kunz, *et. al.* “Negotiating Fragments (Restructuring Quarto Workshop),” Genoa, Italy, Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2018. [Online]. Available:

 GSAPP [ [Link](#) ]  YouTube [ [Link](#) ]  Issuu [ [Link](#) ]  Issuu [ [Link](#) ]  Press [ [Link](#) ]

N. Kunz, “Proportionalities: A Decade of Housing Markets to Subway Stations in Manhattan from 2006 to 2016,” Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

 GSAPP [ [Link](#) ]  GSAPP [ [Link](#) ]  Issuu [ [Link](#) ]

N. Kunz, “Reimaged: John Snow’s Cholera Map of 1854,” Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

 Issuu [ [Link](#) ]

G. S. Coch, S. Nelsen, N. Kunz, K. Warner, “Seeking Refuge: The Geography of Migrant Deaths Along the US-MX Border,” Grad. Sch. of Arch., Plan., and Pres. (GSAPP), Columbia Univ., New York, NY, 2017. [Online]. Available:

 GSAPP [ [Link](#) ]  Medium [ [Link](#) ]

N. Kunz, “Health & Wellness Alliance for Children: Children’s Asthma, Healthy Physical Environments,” Dallas County, Texas, bcWORKSHOP, Dallas, TX, 2014. [Online]. Available:

 Issuu [ [Link](#) ]

N. Kunz, “The Bexar Street Corridor: A Century in Development,” Dallas, Texas, bcWORKSHOP, Dallas, TX, 2013. [Online]. Available:

 Issuu [ [Link](#) ]  Issuu [ [Link](#) ]  bc [ [Link](#) ]  Vimeo [ [Link](#) ]

N. Kunz, “Fenway-Kenmore Square: Comprehensive Planning & Finance,” Cambridge, Massachusetts, Grad. Sch. of Design (GSD), Harvard Univ., Cambridge, MA, 2012. [Online]. Available:

 Issuu [ [Link](#) ]

N. Kunz, “Design Representation: Analog Modeling & Hand Drafting,” Grad. Sch. of Design (GSD), Harvard Univ., Cambridge, MA, 2012. [Online]. Available:

 Issuu [ [Link](#) ]

N. Kunz, “Danelle Plaza: Complete Street Mobility Planning,” Tempe, Arizona, Herberger Inst. for Des. and the Art., Arizona State Univ., Tempe, AZ, 2012. [Online]. Available:

📖 Issuu [ [Link](#) ]

N. Kunz, “Ascension: Multifamily Mixed-Use Real Estate Development Proposal,” Tempe, Arizona, Independent Project, Phoenix, AZ, 2012. [Online]. Available:

📖 Issuu [ [Link](#) ] N. Kunz, “Sui Residence: Second Story Architectural Design,” Paradise Valley, Arizona, Independent Project, Phoenix, AZ, 2011. [Online]. Available:

📖 Issuu [ [Link](#) ]

## Professional Experience

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|--------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Deloitte</b> — Seattle, WA<br>Senior Data Scientist<br><i>Artificial Intelligence</i> [Link]                          | 2021 - 2025 |
| <b>Microsoft</b> — Redmond, WA<br>Data Scientist<br><i>Integrated Visual Augmentation System</i> [Link]                  | 2020 - 2021 |
| <b>Pacific Prospecting Group</b> — Seattle, WA<br>Data Scientist<br><i>Cryptocurrency Mining &amp; Management</i> [Link] | 2017 - 2019 |
| <b>Brawner &amp; Company</b> — Snoqualmie, WA<br>Development Analyst<br><i>Real Estate Development</i> [Link]            | 2016 - 2017 |
| <b>bcWORKSHOP</b> — Dallas, TX<br>Fellow<br><i>Architecture &amp; Planning</i> [Link]                                    | 2013 - 2014 |

## Mentorship Activities

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| <b>Audrey Shin</b> , B.S., Computer Science, Columbia University — New York, NY<br>Machine Learning Engineer Intern, ORO Labs<br><i>Professional development and management in early career engineering.</i>                                    | 2025 - 2026 |
| <b>Albert Rosario-Pichardo</b> , M.S., Urban Planning, Columbia University — New York, NY<br>GIS Analyst Intern, New York City Economic Development Corporation<br><i>Academic and professional development in early career public service.</i> | 2023 - 2024 |
| <b>Albert Ruiz Martinez</b> , B.S., Computer Science, Arizona State University — Tempe, AZ<br>Analyst, Deloitte Consulting<br><i>Professional development and management in early career consulting.</i>                                        | 2023 - 2024 |
| <b>Julia VanPutte</b> , B.S., Operations Research, Cornell University — Ithaca, NY<br>Business Intelligence Engineer Intern, Amazon<br><i>Academic and professional development from undergraduate school to industry.</i>                      | 2022 - 2024 |
| <b>Chawner Hurd</b> , M.B.A., Business, Columbia University — New York, NY<br>Associate Intern, Visa<br><i>Military transition from US Army, 5th Special Forces Group to graduate school and industry.</i>                                      | 2020 - 2023 |
| <b>Michael Bollinger</b> , B.A., Computer Science, Columbia University — New York, NY<br>Software Engineer Intern, Amazon<br><i>Military transition from US Army, 75th Ranger Regiment to undergraduate school and industry.</i>                | 2019 - 2023 |
| <b>Daniel Hineline</b> , B.A., Political Science, Columbia University — New York, NY<br>Associate Intern, Guggenheim Partners<br><i>Military transition from US Army, 82nd Airborne Division to undergraduate school and industry.</i>          | 2018 - 2023 |

## Funding & Support

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### Fellowships & Scholarships

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|-----------------------------------------------------------------------------------------------------|-------------|
| Graduate Engineering Fellowship (\$42,773), Cornell University — Ithaca, NY                         | 2023        |
| Systems Engineering Fellowship (\$31,033), Cornell University — Ithaca, NY                          | 2022        |
| Korean Ancestry Grant (\$8,000), William Orr Dingwall Foundation — Washington, DC                   | 2010 - 2012 |
| Barrett, The Honors College Scholarship (\$4,000), Arizona State University — Tempe, AZ             | 2010 - 2012 |
| AIERF Undergraduate Scholarship (\$1,000), Appraisal Institute — Chicago, IL                        | 2010 - 2011 |
| Study.net Foundation Scholarship (\$500), Study.net — Fort Lauderdale, FL                           | 2010        |
| Herberger Institute for Design & the Arts Scholarship (\$300), Arizona State University — Tempe, AZ | 2010        |
| Center for Global Education Services Scholarship (\$750), Arizona State University — Tempe, AZ      | 2009        |

### Education Benefits & National Service Awards

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|---------------------------------------------------------------------------------------------|-------------|
| Chapter 33: Post-9/11 GI Bill, US Dept. of Veterans Affairs — Washington, DC                | 2022 - 2026 |
| Chapter 31: Veteran Readiness and Employment, US Dept. of Veterans Affairs — Washington, DC | 2017 - 2019 |
| Eli Segal AmeriCorps Education Award, US AmeriCorps — Washington, DC                        | 2014        |

## Honors & Awards

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|-----------------------------------------------------------------------------------------|------|
| Applause Award, Deloitte Consulting — Seattle, WA <a href="#">[Link]</a>                | 2022 |
| Distinguished Alumni Award, Arizona State University — Tempe, AZ <a href="#">[Link]</a> | 2016 |

## Talks & Presentations

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|--------------------------------------------------------------------------------------------------------------|------|
| Applications in Data Science, Cornell Data Science Club, Cornell University — Ithaca, NY                     | 2026 |
| Ground Truth Buffer for Idling, 3-Minutes of Madness, Cornell University — Ithaca, NY <a href="#">[Link]</a> | 2024 |
| Global Geolocated Realtime Data, Flash Talks, Cornell University — Ithaca, NY                                | 2023 |
| Emerging Technology Workshop, Countryside Angels Talks, Ag Ventures Alliance — Mason City, IA                | 2021 |
| Planning Paths in Private Tech, Alumni Panel, Columbia GSAPP — New York, NY <a href="#">[Link]</a>           | 2021 |
| Cloudy Night, End of Year Show, Columbia GSAPP — New York, NY <a href="#">[Link]</a>                         | 2019 |
| Restructuring Quarto Workshop, Municipality of Genoa — Genoa, IT <a href="#">[Link]</a>                      | 2018 |
| The Bexar Street Corridor, US Congresswoman Eddie Bernice Johnson's Office — Dallas, TX                      | 2014 |
| The Bexar Street Corridor, TR Hoover Community Development Corporation — Dallas, TX                          | 2014 |
| The Bexar Street Corridor, bcSHOPFRONT Series, bcWORKSHOP — Dallas, TX <a href="#">[Link]</a>                | 2013 |
| Realizing Interactive Architecture, Honors Symposium, Arizona State University — Tempe, AZ                   | 2012 |

## Media & Press

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|-----------------------------------------------------------------------------------------------------------------|------|
| Urban Planning Semester In Review: Alumni Updates, Columbia GSAPP — New York, NY <a href="#">[Link]</a>         | 2023 |
| Spotlights: Nick Kunz, Cornell Systems Engineering — Ithaca, NY <a href="#">[Link]</a>                          | 2023 |
| Meet the N-K Alumni Nick Kunz, Northwood Anchor, Vol. 126, No. 41, Pg. 4 — Northwood, IA <a href="#">[Link]</a> | 2022 |
| With Special Guest, Nick Kunz, TWA Family Hour Podcast — Seattle, WA <a href="#">[Link]</a>                     | 2020 |
| Restructuring Fourth Workshop, Municipality of Genoa Press — Genoa, IT <a href="#">[Link]</a>                   | 2018 |
| Achievement Oriented: Nick Kunz, ASU Magazine, Vol. 20, No. 2, Pg. 7 — Tempe, AZ                                | 2016 |
| Welcome Nick Kunz, bcWORKSHOP — Dallas, TX                                                                      | 2013 |

## Academic Service

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|---------------------------------------------------------------------------------|-------------|
| Tuesday Talks Mentorship, Columbia GSAPP — New York, NY <a href="#">[Link]</a>  | 2026        |
| Alumni Mentorship Program, Columbia GSAPP — New York, NY <a href="#">[Link]</a> | 2023 - 2024 |

## Military Service

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### Assignments

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|----------------------------------------------------------------------------------------------------------|-------------|
| US Army, 2nd Ranger Battalion, 75th Ranger Regiment — Fort Lewis, WA<br><i>Infantry</i>                  | 2015 - 2016 |
| US Army, Ranger Training Detachment — Fort Benning, GA<br><i>Ranger Assessment and Selection Program</i> | 2015        |

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|---------------------------------------------------------------------------------------------------------------------------|------|
| <b>US Army, 1st Special Warfare Training Group</b> — Fort Bragg, NC<br><i>Special Forces Assessment and Selection</i>     | 2014 |
| <b>US Army, 1st Special Warfare Training Group</b> — Fort Bragg, NC<br><i>Special Forces Preparation and Conditioning</i> | 2014 |
| <b>US Army, 507th Parachute Infantry Regiment</b> — Fort Benning, GA<br><i>Airborne School</i>                            | 2014 |
| <b>US Army, 19th Infantry Regiment</b> — Fort Benning, GA<br><i>Infantry School</i>                                       | 2014 |

Decorations

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|---------------------------------------------------------------------------------------------------------|------|
| <b>Expert Badge, Rifle Component Bar</b> , US Army, 75th Ranger Regiment — Fort Lewis, WA               | 2016 |
| <b>Global War on Terrorism Service Medal</b> , US Army, 75th Ranger Regiment — Fort Lewis, WA           | 2016 |
| <b>Driver Badge, Wheeled Vehicle Component Bar</b> , US Army, 75th Ranger Regiment — Fort Lewis, WA     | 2015 |
| <b>Parachutist Badge, Unit Trimming</b> , US Army, 507th Parachute Infantry Regiment — Fort Benning, GA | 2014 |
| <b>Infantry Blue Chord Fourragère</b> , US Army, 19th Infantry Regiment — Fort Benning, GA              | 2014 |
| <b>National Defense Service Medal</b> , US Army, 19th Infantry Regiment — Fort Benning, GA              | 2014 |
| <b>Army Service Ribbon</b> , US Army, 19th Infantry Regiment — Fort Benning, GA                         | 2014 |

Civil Service

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|-------------------------------------------------------------------------------------------|-------------|
| <b>AmeriCorps</b> — Dallas, TX<br><i>Volunteer in Service to America (VISTA)</i>          | 2013 - 2014 |
| <b>Open Architecture Collaborative</b> — Dallas, TX<br><i>Architecture &amp; Planning</i> | 2013 - 2014 |
| <b>Trinity Park Conservancy</b> — Dallas, TX<br><i>Public Park Conservation</i>           | 2013 - 2014 |

Professional Memberships

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|-------------------------------------------------------------------------------|----------------|
| <b>Sigma Xi</b> — Research Triangle Park, NC                                  | 2026 - Present |
| <b>The New York Academy of Sciences</b> — New York, NY                        | 2024 - Present |
| <b>AmeriCorps Alums of North Texas</b> — Dallas, TX                           | 2014 - Present |
| <b>Barrett, The Honors College Alumni Association</b> — Tempe, AZ             | 2012 - Present |
| <b>US Military Veterans of Columbia University</b> — New York, NY             | 2017 - 2018    |
| <b>Architecture for Humanity, Dallas/Fort Worth Chapter</b> — Dallas, TX      | 2013 - 2014    |
| <b>Social Economic Environmental Design Network</b> — Princeton, NJ           | 2013 - 2014    |
| <b>Urban Land Institute Arizona</b> — Phoenix, AZ                             | 2011 - 2012    |
| <b>United States Green Building Council, Central AZ Chapter</b> — Phoenix, AZ | 2009 - 2012    |
| <b>American Institute of Architecture Students</b> — Washington, DC           | 2009 - 2010    |

Technical Skills

Languages

**Scripting:** Python, Bash  
**Statistical:** R, Stata  
**Compiled:** C/C++, Fortran  
**Database:** SQL, NoSQL, Cypher  
**Typesetting:** L<sup>A</sup>T<sub>E</sub>X, Markdown

Frameworks

**Data Science:** NumPy, SciPy, Pandas, statsmodels, Featuretools, Tidyverse  
**Visualization:** Matplotlib, Seaborn, Plotly, Processing, Shiny, ggplot  
**Statistical Learning:** Scikit-Learn, XGBoost, LightGBM, CatBoost  
**Reinforcement Learning:** OpenAI Baselines, Stable Baselines  
**AI Engineering:** Pinecone, Langchain, LangGraph  
**Deep Learning:** Pytorch, TensorFlow, ONNX

## Deployment

**DevOps:** Git, Ubuntu, Docker, CI/CD

**Databases:** PostgreSQL, SQLite, CosmosDB, Neo4j

**Web Services:** Flask, Gunicorn, Nginx

**Web Automation:** Selenium, Puppeteer

**Cloud Platforms:** Azure, AWS, Databricks

**Security:** STIGs, NIST, ZAP

## Software

**Development Environments:** VS Code, RStudio, Jupyter

**Geographic Information Systems:** ArcGIS, QGIS, OSM, Leaflet, Mapbox

**3D Modeling & Optimization:** Rhino, Grasshopper, SketchUp, AutoCAD, Unity

**Design & Illustration:** Adobe Photoshop, Illustrator, InDesign

**Financial Modeling:** Excel, Macros, CoStar

## Prototyping

**Project Management:** Agile, Scrum, Critical Path

**Sensors & Hardware:** Arduino, Raspberry Pi

**Electrical:** Soldering, Wiring, Safety

**Analogs:** Drawing, Hand Drafting, Physical Modeling

## Personal Activities

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### Music & Live Audio

**Atlas North American Tour** — US & Canada

2013

The Word Alive (*Fearless Records*), Co-Tour Manager

The Word Alive (*Fearless Records*), Guitar Tech

**Fight the Silence Tour** — US

2013

The Word Alive (*Fearless Records*), Guitar Tech

For Today (*Razor & Tie Records*), Guitar Tech

**All Stars Tour** — US

2012

The Word Alive (*Fearless Records*), Guitar Tech

A Skylit Drive (*Fearless Records*), Guitar Tech

**Van's Warped Tour** — US & Canada

2011

The Word Alive (*Fearless Records*), Co-Tour Manager

Of Mice & Men (*Rise Records*), Guitar Tech

**Regional Performances** — US

2011

The Rendering (*Authentik Artists*), Guitar